

Improper integrals – 1

Convergence: the two scales and comparison

12 сентября 2026 года

Theoretical minimum.

Definition. $\int_a^\omega f = \lim_{b \rightarrow \omega} \int_a^b f$, where ω is either $+\infty$ or a point near which f is unbounded. An integral with several singular points is split so that each piece has exactly one, and it converges only when every piece does: the two ends are never allowed to compensate each other (that is a different object, the principal value, see Improper integrals – 4).

Theorem 1 (the two scales). At infinity $\int_a^\infty \frac{dx}{x^p}$ converges exactly for $p > 1$ and $\int_a^\infty \frac{dx}{x \ln^p x}$ exactly for $p > 1$; at a singular point $\int_c \frac{dx}{(x-c)^p}$ converges exactly for $p < 1$ and $\int_c \frac{dx}{(x-c) |\ln(x-c)|^p}$ exactly for $p > 1$. Comparison and limit comparison transfer convergence for eventually one-signed integrands, and only for them.

Theorem 2 (Cauchy criterion). $\int_a^\infty f$ converges if and only if for every $\varepsilon > 0$ there is B with $|\int_{b_1}^{b_2} f| < \varepsilon$ for all $b_2 > b_1 > B$. Absolute convergence implies convergence; the converse fails, and the whole of Improper integrals – 2 is about that failure.

Theorem 3 (dyadic decomposition). For $f \geq 0$ non-increasing on $[1, \infty)$ the integral $\int_1^\infty f$, the series $\sum_{n \geq 1} f(n)$ and the series $\sum_{k \geq 0} 2^k f(2^k)$ converge or diverge together. The first equivalence is the integral test (proved in Series and infinite products – 1), the second is Cauchy condensation; monotonicity is essential for both.

Fact 1 (what convergence does not give). Convergence of $\int_0^\infty f$ does not force $f(x) \rightarrow 0$ and does not even force f to be bounded. It does force it if f is uniformly continuous, and if $f \geq 0$ is non-increasing it forces the much stronger $xf(x) \rightarrow 0$. Without monotonicity one can only say $\liminf xf(x) = 0$.

-
1. Prove both scales of Theorem 1 together with their logarithmic refinements, prove Theorem 3, and deduce that there is no slowest convergent integral: for every convergent $\int_1^\infty f$ with $f > 0$ there is a convergent $\int_1^\infty g$ with $g/f \rightarrow \infty$.
 2. (МФТИ) Investigate the convergence of the improper integral $\int_0^{+\infty} \frac{x - \sin x}{x^\alpha} dx$ for all values of the parameter $\alpha \in \mathbb{R}$.
 3. (VJIMC 2012) Let $f: [1, \infty) \rightarrow (0, \infty)$ be a non-increasing function such that $\limsup_{n \rightarrow \infty} \frac{f(2^{n+1})}{f(2^n)} < \frac{1}{2}$. Prove that $\int_1^\infty f(x) dx < \infty$.
 4. (IMS 2022) Let $f: (0, +\infty) \rightarrow (0, +\infty)$ be integrable on every bounded interval and suppose that $\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} = 1$. Prove that for every $\alpha < 1$

$$\int_1^\infty \frac{f(x)}{x^\alpha} dx = \infty.$$

5. (RMO) Let $a \in (1, \infty)$ and let $f: [0, \infty) \rightarrow \mathbb{R}$ be continuous with $\lim_{x \rightarrow \infty} xf(x) \in \mathbb{R}$.
 - a) Show that $\int_1^\infty \frac{f(x)}{x} dx$ and $\lim_{t \rightarrow \infty} t \int_1^a f(xt) dx$ both exist, are finite and are equal.
 - b) Calculate $\lim_{t \rightarrow \infty} t \int_1^a \frac{dx}{1+x^t}$.